Chapter 6. Metrization Theorems and Paracompactness

Local finiteness, Nagata–Smirnov, paracompactness, Smirnov metrization

Contents

6.1 Local finiteness **2**

6.2 The Nagata–Smirnov metrization theorem **2**

6.3 Paracompactness **4**

6.4 The Smirnov metrization theorem **6**

6.5 Selected exercises **6**

6.1 Local finiteness

Motivation. Locally finite families replace the countable basis, since piecewise definitions over them behave well.

A collection $\mathcal{A}$ of subsets of X is **locally finite** if every point has a neighbourhood meeting only finitely many elements of $\mathcal{A}$; it is **countably locally finite** (σ-locally finite) if it is a countable union of locally finite collections. A collection $\mathcal{B}$ **refines** $\mathcal{A}$ (is a **refinement** of $\mathcal{A}$) if each $B \in \mathcal{B}$ lies in some $A \in \mathcal{A}$; it is an open (closed) refinement if its elements are open (closed).

Lemma 6.1 (39.1)

Let $\mathcal{A}$ be locally finite.
(a) Every subcollection of $\mathcal{A}$ is locally finite.
(b) $\{\overline{A} : A \in \mathcal{A}\}$ is locally finite.
(c) $\overline{\bigcup_{A\in\mathcal{A}} A} = \bigcup_{A\in\mathcal{A}} \overline{A}$.

Proof. (a) is clear.

(b) An open set U meeting $\overline{A}$ meets A; so a neighbourhood meeting finitely many A meets only the corresponding $\overline{A}$.

(c) The inclusion $\supset$ always holds. Let $x \in \overline{\bigcup A}$ and let U be a neighbourhood of x meeting only $A_1, \dots, A_k$. If $x \notin \overline{A_1} \cup \cdots \cup \overline{A_k}$, then $U - (\overline{A_1} \cup \cdots \cup \overline{A_k})$ is a neighbourhood of x missing every element of $\mathcal{A}$, a contradiction. ■

Without local finiteness (c) fails: the one-point sets $\{1/n\}$ have union with closure containing 0.

Lemma 6.2 (39.2)

If X is metrizable, every open covering $\mathcal{A}$ of X has a countably locally finite open refinement $\mathcal{E}$ that covers X.

Proof. Fix a metric d and a well-ordering $<$ of $\mathcal{A}$ (Theorem 1.25). For $n \in \mathbb{Z}_+$ and $U \in \mathcal{A}$ let

$$S_n(U) = \{x : B(x, 1/n) \subset U\}, \qquad T_n(U) = S_n(U) - \bigcup_{V<U} V, \qquad E_n(U) = \bigcup_{x\in T_n(U)} B(x, 1/3n).$$

If $V < U$, $x \in T_n(V)$, $y \in T_n(U)$, then $B(x, 1/n) \subset V$ and $y \notin V$, so $d(x, y) \geq 1/n$. Hence distinct sets $E_n(U), E_n(V)$ are at distance at least $1/3n$, a ball of radius $1/6n$ meets at most one of them, and $\mathcal{E}_n = \{E_n(U)\}_U$ is locally finite. Also $E_n(U) \subset U$, since $B(x, 1/3n) \subset B(x, 1/n) \subset U$ for $x \in T_n(U)$. Finally, given x, let U be the first element of $\mathcal{A}$ containing x and choose n with $B(x, 1/n) \subset U$: then $x \in S_n(U)$ and x lies in no $V < U$, so $x \in T_n(U) \subset E_n(U)$. Take $\mathcal{E} = \bigcup_n \mathcal{E}_n$. ■

6.2 The Nagata–Smirnov metrization theorem

Motivation. A space is metrizable if and only if it is regular with a countably locally finite basis.

A set is a G_δ **set** if it is a countable intersection of open sets. In a metric space every closed set A is G_δ: $A = \bigcap_n \{x : d(x, A) < 1/n\}$.

Lemma 6.3 (40.1)

Let X be regular with a countably locally finite basis $\mathcal{B}$. Then X is normal and every closed set is a G_δ set.

Proof.

Step 1. for each open W there are open $U_1, U_2, \dots$ with $W = \bigcup U_n = \bigcup \overline{U_n}$. Write $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite, let $\mathcal{C}_n = \{B \in \mathcal{B}_n : \overline{B} \subset W\}$ (locally finite) and $U_n = \bigcup_{B \in \mathcal{C}_n} B$. By Lemma 6.1(c), $\overline{U_n} = \bigcup_{\mathcal{C}_n} \overline{B} \subset W$. If $x \in W$, regularity gives a basis element $B \ni x$ with $\overline{B} \subset W$; $B \in \mathcal{C}_n$ for some n, so $x \in U_n$.

Step 2. if C is closed, apply Step 1 to $W = X - C$: $C = \bigcap_n (X - \overline{U_n})$, a G_δ.

Step 3. for disjoint closed C, D, Step 1 with $W = X - D$ gives open U_n covering C with closures missing D, and with $W = X - C$ open V_n covering D with closures missing C. Now finish exactly as in the proof of Theorem 4.6. ■

Lemma 6.4 (40.2)

If X is normal and A is a closed G_δ set, there is a continuous $f : X \to [0, 1]$ with $f = 0$ on A and $f > 0$ off A.

Proof. Write $A = \bigcap U_n$ with U_n open. Urysohn gives $f_n : X \to [0, 1]$ equal to 0 on A and 1 on $X - U_n$. Let $f = \sum_n f_n / 2^n$, continuous by uniform convergence (Theorem 2.42). It vanishes on A; if $x \notin A$, then $x \notin U_n$ for some n, so $f(x) \geq 2^{-n} > 0$. ■

Theorem 6.5 (40.3, Nagata–Smirnov metrization theorem)

X is metrizable iff X is regular and has a countably locally finite basis.

Proof. ($\Leftarrow$)

Step 1. let $\mathcal{B} = \bigcup \mathcal{B}_n$ with $\mathcal{B}_n$ locally finite. X is normal and $X - B$ is a closed G_δ for each $B \in \mathcal{B}$ (Lemma 6.3), so Lemma 6.4 gives $f_{n,B} : X \to [0, 1/n]$ positive on B and zero off B. For $x_0 \in U$ choose $B \in \mathcal{B}_n$ with $x_0 \in B \subset U$; then $f_{n,B}(x_0) > 0$ and $f_{n,B} = 0$ off U.

Step 2. let $J = \{(n, B) : B \in \mathcal{B}_n\}$ and $F(x) = (f_{n,B}(x)) \in [0, 1]^J$; give $[0, 1]^J$ the uniform metric $\bar{\rho}$. By the proof of Theorem 4.13, F is injective and maps open sets of X to sets open in $F(X)$ for the product topology, hence also for the finer uniform topology (Theorem 2.35).

Step 3 (F is continuous into the uniform metric). let $x_0 \in X$ and $\epsilon > 0$. For each n choose a neighbourhood U_n of x_0 meeting only finitely many elements of $\mathcal{B}_n$; all but finitely many $f_{n,B}$ vanish on U_n, and for the remaining ones continuity gives a smaller neighbourhood W_n on which $|f_{n,B}(x) - f_{n,B}(x_0)| \leq \epsilon/2$ for every $B \in \mathcal{B}_n$. Choose N with $1/N \leq \epsilon/2$ and let $W = W_1 \cap \cdots \cap W_N$. For $x \in W$ the coordinates with $n \leq N$ differ by at most $\epsilon/2$, and those with $n > N$ lie in $[0, 1/n]$, so differ by at most $1/n < \epsilon/2$. Thus $\bar{\rho}(F(x), F(x_0)) \leq \epsilon/2 < \epsilon$, and F imbeds X in a metric space.

($\Rightarrow$) A metrizable space is regular (Theorem 4.7). For each m let $\mathcal{A}_m$ be the covering by balls of radius $1/m$ and $\mathcal{D}_m$ a countably locally finite open refinement covering X (Lemma 6.2); its elements have diameter at most $2/m$. $\mathcal{D} = \bigcup_m \mathcal{D}_m$ is countably locally finite. It is a basis: if $x \in U$ open, choose m with $B(x, 1/m) \subset U$ and $D \in \mathcal{D}_{3m}$ containing x; D has diameter at most $2/3m < 1/m$, so $D \subset B(x, 1/m) \subset U$. ■

6.3 Paracompactness

Motivation. Paracompactness, every open cover has a locally finite refinement, gives partitions of unity for arbitrary covers, and metric spaces have it.

X is **paracompact** if every open covering has a locally finite open refinement that covers X. Compact spaces are paracompact, and so is $\mathbb{R}^n$ (Theorem 6.9 below).

Theorem 6.6 (41.1)

Every paracompact Hausdorff space is normal.

Proof. Regularity: let $a \notin B$, B closed. For each $b \in B$ choose open $U_b \ni b$ with $a \notin \overline{U_b}$ (Hausdorff). Cover X by the U_b and $X - B$, take a locally finite open refinement $\mathcal{C}$ covering X, and let $\mathcal{D}$ be the elements of $\mathcal{C}$ meeting B. Each $D \in \mathcal{D}$ is not inside $X - B$, so $D \subset U_b$ for some b and $a \notin \overline{D}$. $V = \bigcup_{\mathcal{D}} D$ is open, contains B, and $\overline{V} = \bigcup \overline{D} \not\ni a$ (Lemma 6.1). Then V and $X - \overline{V}$ separate B and a. Normality: the same argument with a closed set A in place of a, using regularity to choose U_b with $\overline{U_b} \cap A = \emptyset$. ■

Theorem 6.7 (41.2)

Every closed subspace of a paracompact space is paracompact.

Proof. Let Y be closed and $\mathcal{A}$ a covering of Y by sets open in Y; for each $A \in \mathcal{A}$ choose A' open in X with $A' \cap Y = A$. Cover X by the A' together with $X - Y$, and take a locally finite open refinement $\mathcal{B}$ covering X. Then $\{B \cap Y : B \in \mathcal{B}\}$ covers Y, is locally finite, consists of sets open in Y, and refines $\mathcal{A}$ (a B inside $X - Y$ gives the empty set). ■

A paracompact space can have non-paracompact subspaces: $\overline{S}_\Omega$ is compact, but S_Ω is not paracompact (it is not metrizable yet is locally metrizable; see Theorem 6.14). The product $\mathbb{R}^2_\ell$ of paracompact spaces is not paracompact, because it is Hausdorff but not normal.

Lemma 6.8 (41.3, Michael)

Let X be regular. The following are equivalent: every open covering of X has a refinement that is (1) an open covering of X and countably locally finite;
(2) a covering of X and locally finite;
(3) a closed covering of X and locally finite;
(4) an open covering of X and locally finite.

Proof. (4)⇒(1) is trivial.

(1)⇒(2): let $\mathcal{B} = \bigcup \mathcal{B}_n$ be an open refinement as in (1), $V_i = \bigcup_{U \in \mathcal{B}_i} U$, and for $U \in \mathcal{B}_n$ put $S_n(U) = U - \bigcup_{i<n} V_i$. The sets $S_n(U)$ refine $\mathcal{A}$ and cover X: for x, take the smallest N with x in some $U \in \mathcal{B}_N$; then $x \in S_N(U)$. Local finiteness: with $x \in U \in \mathcal{B}_N$, the open set $U \subset V_N$ meets no $S_n(V)$ with $n > N$; for each $n \le N$ choose a neighbourhood W_n meeting finitely many elements of $\mathcal{B}_n$, hence finitely many $S_n(V)$. Then $U \cap W_1 \cap \cdots \cap W_N$ meets finitely many sets $S_n(V)$.

(2)⇒(3): let $\mathcal{B}$ be the open sets U with $\overline{U}$ contained in some element of $\mathcal{A}$; it covers X by regularity. Take a locally finite refinement $\mathcal{C}$ covering X; $\{\overline{C}\}$ is locally finite (Lemma 6.1), closed, covers X and refines $\mathcal{A}$.

(3)⇒(4): let $\mathcal{B}$ be a locally finite refinement of $\mathcal{A}$ covering X (by (3)). Let $\mathcal{C}$ be the open sets meeting only finitely many elements of $\mathcal{B}$; it covers X. By (3) take a closed locally finite refinement $\mathcal{D}$ of $\mathcal{C}$ covering X; each $D \in \mathcal{D}$ meets only finitely many elements of $\mathcal{B}$. For $B \in \mathcal{B}$ let $C(B) =$

$X - \bigcup\{D \in \mathcal{D} : D \cap B = \emptyset\}$, open by Lemma 6.1(c) and containing B, and choose $F(B) \in \mathcal{A}$ with $B \subset F(B)$. The collection $\mathcal{E} = \{C(B) \cap F(B) : B \in \mathcal{B}\}$ is an open refinement of $\mathcal{A}$ covering X. It is locally finite: a point has a neighbourhood W meeting only $D_1, \ldots, D_k \in \mathcal{D}$, so $W \subset D_1 \cup \cdots \cup D_k$. If D meets $C(B)$, then D meets B (otherwise $D \subset X - C(B)$); hence each D_i meets only finitely many $C(B)$, and so does W. ■

Theorem 6.9 (41.4)

Every metrizable space is paracompact.

Proof. It is regular, and condition (1) of Lemma 6.8 holds by Lemma 6.2. ■

Theorem 6.10 (41.5)

Every regular Lindelöf space is paracompact.

Proof. A countable subcovering is a countable union of one-element (hence locally finite) collections, so condition (1) of Lemma 6.8 holds. ■

So $\mathbb{R}_\ell$ is paracompact, while $\mathbb{R}_\ell^2$ is not; and $\mathbb{R}^\omega$ in the product topology is paracompact (metrizable), whereas $\mathbb{R}^J$ for uncountable J is not (it is not normal).

Lemma 6.11 (41.6)

Let X be paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ an indexed open covering. There is a locally finite indexed open covering $\{V_\alpha\}_{\alpha \in J}$ with $\overline{V_\alpha} \subset U_\alpha$ for each α (a **precise refinement**).

Proof. Let $\mathcal{A}$ be the open sets A with $\overline{A}$ in some U_α; it covers X by regularity (Theorem 6.6). Take a locally finite open refinement $\mathcal{B}$ covering X, and for each B choose $f(B) \in J$ with $\overline{B} \subset U_{f(B)}$. Let $V_\alpha = \bigcup\{B : f(B) = \alpha\}$. The V_α are open and cover X; a neighbourhood meeting finitely many B meets finitely many V_α; and $\overline{V_\alpha} = \bigcup_{f(B)=\alpha} \overline{B} \subset U_\alpha$ by Lemma 6.1(c). ■

Theorem 6.12 (41.7)

If X is paracompact Hausdorff and $\{U_\alpha\}_{\alpha \in J}$ is an indexed open covering, there is a partition of unity dominated by $\{U_\alpha\}$: continuous $\phi_\alpha : X \to [0,1]$ with $\operatorname{supp} \phi_\alpha \subset U_\alpha$, $\{\operatorname{supp} \phi_\alpha\}$ locally finite, and $\sum_\alpha \phi_\alpha = 1$.

Proof. By Lemma 6.11 twice, take locally finite open coverings $\{V_\alpha\}$, $\{W_\alpha\}$ with $\overline{W_\alpha} \subset V_\alpha$, $\overline{V_\alpha} \subset U_\alpha$. Urysohn (with X normal) gives ψ_α equal to 1 on $\overline{W_\alpha}$ and 0 off V_α, so $\operatorname{supp} \psi_\alpha \subset \overline{V_\alpha} \subset U_\alpha$, and these supports are locally finite (Lemma 6.1(b)). $\Psi = \sum \psi_\alpha$ is near each point a finite sum of continuous functions, hence continuous, and positive since the W_α cover. Put $\phi_\alpha = \psi_\alpha / \Psi$. ■

Theorem 6.13 (41.8)

Let X be paracompact Hausdorff, $\mathcal{C}$ a locally finite collection of subsets, and $\epsilon_C > 0$ for each $C \in \mathcal{C}$. There is a continuous $f : X \to \mathbb{R}$ with $f > 0$ everywhere and $f(x) \le \epsilon_C$ for $x \in C$.

Proof. Cover X by open sets each meeting finitely many elements of $\mathcal{C}$, and take a dominated partition of unity $\{\phi_U\}$ indexed by this covering. Let δ_U be the minimum of ϵ_C over the finitely many C meeting U ($\delta_U = 1$ if there are none), and $f = \sum_U \delta_U \phi_U$, continuous and positive. If $x \in C$ and $\phi_U(x) > 0$, then $x \in U$, so U meets C and $\delta_U \le \epsilon_C$; hence $f(x) \le \epsilon_C \sum \phi_U(x) = \epsilon_C$. ■

6.4 The Smirnov metrization theorem

Motivation. A paracompact Hausdorff space is metrizable exactly when it is locally metrizable.

X is **locally metrizable** if each point has a neighbourhood that is metrizable in the subspace topology.

Theorem 6.14 (42.1, Smirnov metrization theorem)

X is metrizable iff it is a paracompact Hausdorff space that is locally metrizable.

Proof. ($\Rightarrow$) Theorem 6.9.

($\Leftarrow$) X is regular (Theorem 6.6); we find a countably locally finite basis and apply Theorem 6.5. Cover X by open metrizable sets and take a locally finite open refinement $\mathcal{C}$ covering X; each $C \in \mathcal{C}$ has a metric d_C, and the balls $B_C(x,\epsilon) = \{y \in C : d_C(x,y) < \epsilon\}$ are open in X. For each m, let $\mathcal{A}_m$ be the covering by all balls $B_C(x, 1/m)$ ($C \in \mathcal{C}$, $x \in C$), and $\mathcal{D}_m$ a locally finite open refinement covering X. Then $\mathcal{D} = \bigcup \mathcal{D}_m$ is countably locally finite. Basis: let $x \in U$ open. Only finitely many $C_1, \dots, C_k \in \mathcal{C}$ contain x; choose ϵ_i with $B_{C_i}(x,\epsilon_i) \subset U \cap C_i$ and m with $2/m < \min \epsilon_i$. Choose $D \in \mathcal{D}_m$ with $x \in D$; then $D \subset B_C(y, 1/m)$ for some C and y, and $x \in C$, so $C = C_i$ for some i. By the triangle inequality $D \subset B_{C_i}(x, 2/m) \subset U$. ■

6.5 Selected exercises

Motivation. Exercises from Munkres on why countable local finiteness is needed, a product theorem for paracompactness, and non-paracompactness of S_Ω.

Exercise 6.1 (§39, Ex. 5; countably locally finite means countable)

Let X have a countable basis. Show that a collection $\mathcal{A}$ of subsets of X is countably locally finite iff it is countable.

Solution.

($\Leftarrow$) A countable $\mathcal{A} = \{A_1, A_2, \dots\}$ is the countable union of the collections $\{A_n\}$, each of which is finite, hence locally finite.

($\Rightarrow$) It suffices to show that a locally finite collection $\mathcal{A}_0$ is countable. Let $\mathcal{B}$ be a countable basis.

Step 1. Let $\mathcal{B}' \subset \mathcal{B}$ consist of the basis elements meeting only finitely many elements of $\mathcal{A}_0$. $\mathcal{B}'$ covers X: each x has a neighbourhood W meeting finitely many elements of $\mathcal{A}_0$, and a basis element B with $x \in B \subset W$ lies in $\mathcal{B}'$.

Step 2. Every nonempty $A \in \mathcal{A}_0$ contains a point, which lies in some $B \in \mathcal{B}'$; so A meets B. Hence $\mathcal{A}_0 - \{\emptyset\} \subset \bigcup_{B \in \mathcal{B}'} \{A \in \mathcal{A}_0 : A \cap B \neq \emptyset\}$, a countable union of finite sets, and $\mathcal{A}_0$ is countable (Theorem 1.15). ■

Exercise 6.2 (§40, Ex. 3; locally finite bases)

Show that a T_1 space X with a locally finite basis $\mathcal{B}$ is discrete.

Solution. Let $x \in X$.

Step 1. Some neighbourhood W of x meets only finitely many elements of $\mathcal{B}$. Every basis element containing x meets W, so there are only finitely many, say $B_1, \dots, B_k$ ($k \geq 1$ since $\mathcal{B}$ covers X). Then $O = B_1 \cap \cdots \cap B_k$ is an open set containing x.

Step 2. $O = \{x\}$. Let $y \neq x$. Since $\{y\}$ is closed, $X - \{y\}$ is a neighbourhood of x, so some $B \in \mathcal{B}$ has $x \in B \subset X - \{y\}$. This B is one of the B_i, so $y \notin O$.

Hence every one-point set is open and X is discrete. ■

Exercise 6.3 (§41, Ex. 2; products with compact spaces)

(a) Show that if X is paracompact and Y is compact, then $X \times Y$ is paracompact.
(b) Conclude that S_Ω is not paracompact.

Solution.

(a) Let $\mathcal{A}$ be an open covering of $X \times Y$.

Step 1. For $x \in X$, the compact slice $x \times Y$ is covered by finitely many $A_{x,1}, \dots, A_{x,k_x} \in \mathcal{A}$. By the tube lemma (Lemma 3.21) there is an open $W_x \ni x$ with $W_x \times Y \subset A_{x,1} \cup \cdots \cup A_{x,k_x}$.

Step 2. $\{W_x\}$ covers X; let $\mathcal{C}$ be a locally finite open refinement covering X, and for each $C \in \mathcal{C}$ choose $x(C)$ with $C \subset W_{x(C)}$. Let

$$\mathcal{D} = \{(C \times Y) \cap A_{x(C),i} : C \in \mathcal{C},\ 1 \leq i \leq k_{x(C)}\}.$$

Its elements are open and each lies in an element of $\mathcal{A}$.

Step 3. $\mathcal{D}$ covers $X \times Y$: if $x \in C \in \mathcal{C}$, then $x \times Y \subset C \times Y \subset W_{x(C)} \times Y \subset \bigcup_i A_{x(C),i}$, so each point of $x \times Y$ lies in some $(C \times Y) \cap A_{x(C),i}$.

Step 4. $\mathcal{D}$ is locally finite: for (x, y), choose a neighbourhood N of x meeting only finitely many $C \in \mathcal{C}$. Then $N \times Y$ meets only the elements of $\mathcal{D}$ coming from these C, finitely many for each.

(b) $\overline{S}_\Omega$ is compact: it is a closed interval of a well-ordered set, which has the least upper bound property (the set of upper bounds of a bounded nonempty subset has a least element), so Theorem 3.24 applies. If S_Ω were paracompact, then $S_\Omega \times \overline{S}_\Omega$ would be paracompact by (a), and Hausdorff, hence normal by Theorem 6.6. This contradicts Example 4.3(b). ■